Data Structures

Course Code	Course Name	Credits
IFM4114	Data Structures	04

Contact Hours			Credits Assigned			
Theory	**Practical**	**Tutorial**	**Theory**	**Practical**	**Tutorial**	**Total**
03	-	01	03	-	01	04

	Theory					Term Work / Practical/Oral			
Internal Assessment				**End Sem Exam**	**Duration Of End Sem Exam**	**Term Work**	**Pract.**	**Oral**	**Total**
Test	**Continuous Evaluation**	**Attendance**	**Total Internal**						
15	**10**	**05**	**30**	**70**	**3 Hours**	**-**	**-**	**-**	**100**

Course outcome

1. Explain various types of data structures and analyze their computational complexity.
2. Implement linear data structures and perform operations such as insertion, deletion, searching, and traversal.
3. Apply stack and queue concepts to solve real-world problems such as expression evaluation and recursion.
4. Construct and manipulate trees and graphs, and apply traversal algorithms.
5. Select appropriate data structures to design efficient solutions for computational problems.
6. Design and implement database-driven applications by integrating database design principles, SQL programming, and transaction management to solve real-world problems.

Course Objectives

1. To understand the fundamental concepts of data structures and their role in efficient problem solving.
2. To analyze and evaluate the performance of algorithms using time and space complexity.
3. To implement linear data structures such as arrays, linked lists, stacks, and queues.
4. To design and apply non-linear data structures such as trees and graphs.
5. To develop problem-solving skills using appropriate data structures and algorithmic techniques
6. Apply searching and sorting algorithms to optimize data processing and improve computational efficiency.

Detailed Syllabus

Module / Unit	Course Module / Contents		Hours	Marks Weightage
	Module I : Introduction		**08**	**20%**
1	1.1	Introduction to Data structures, Algorithm Design paradigms - motivation,		
	1.2	Concept of algorithmic efficiency, run time analysis of algorithms, Asymptotic Notations.		
	1.3	Recurrences- substitution method, recursion tree method, master method		
	Module II : Data Structures Types		**09**	**20%**
2	2.1	Arrays- Implementation and analysis of basic operations of array. Stack-implementation and analysis of push and pop operations,		
	2.2	infix, postfix, and prefix.		
	2.3	Queue- Implementation and analysis of basic operations of Queue,		
	2.4	Circular Queue and its analysis.		
3	**Module III : Linked Lists and Trees**		**09**	**15%**
	3.1	Insertion and deletion of a node to single linked list, doubly linked list with basic operations and analysis.		
	3.2	Binary search tree- Operations on Binary search tree. Implementation of B.S.T. Constructing a binary search tree based on its given infix and pre/post fix order.		
	3.3	AVL Tree- Constructing an AVL Tree, Insertion and Deletion of a node, searching a node.		
4	**Module IV : Graph and Their Applications**		**09**	**15%**
	4.1	Introduction, Sequential Representation of Graph (Adjacency and Path Matrix), Warshall Algorithms, Linked Representation of Graph,		
	4.2	Different Operations on Graphs, Traversing A Graph using DFS and BFS, Spanning Trees-Introduction. Representation of Spanning tree, Constructing A Spanning Tree, Prim's and Kruskal Algorithm.		
5	**Module V : Algorithmic Paradigm**		**09**	**15%**
	5.1	Divide and conquer. Structure of divide-and-conquer algorithms: examples; Binary search, quick sort, Merge sort, Heap Sort,Strassen Multiplication; Analysis of divide and conquer run time recurrence relations.		
	5.2	Greedy Method Fractional Knapsack problem, Single source shortest paths, Job sequencing problem, Huffman Coding.		
6	**Module VI: Dynamic Programming**		**08**	**15%**
	6.1	Dynamic programming difference between dynamic programming and divide and conquer,		
	6.2	Applications: chain Matrix multiplication, Traveling salesman Problem, knapsack problem, All pair shortest path algorithm		

Total			**52**	**100%**

Recommended Books:

No.	Book Title	Author(s)	Publisher	ISBN-13 / Edition
1	Mastering Data Structures Through C Language	J.B. Dixit	Laxmi Publications	978-8131804476 (1st Ed.)
2	Data Structures Using C	Yashavant Kanetkar	BPB Publications	978-8183334617 (2nd Ed.)
3	Introduction to Algorithms	T. H. Cormen, C. E. Leiserson, R. L. Rivest, & C. Stein	MIT Press / PHI	978-0262046305 (4th Ed.)
4	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni, & S. Rajasekaran	Galgotia Publications	978-8175152571 (2nd Ed.)
5	Data Structures Through C Language	S. Chattopadhyay, D. Ghoshdastidar, & M. Chattopadhyay	BPB Publications	978-8176563611 (1st Ed.)
6	Data Structures and Program Design in C	Robert L. Kruse, Bruce P. Leung, & Clovis L. Tondo	Pearson Education	978-0132883665 (2nd Ed.)

DBMS & Data Structures Lab

Course Code	Course Name	Credits
IFM4116	DBMS & Data Structures Lab	02

Contact Hours			Credits Assigned			
Theory	**Practical**	**Tutorial**	**Theory**	**Practical**	**Tutorial**	**Total**
-	04	-	-	02	-	02

	Theory					Term Work / Practical/Oral			
Internal Assessment				**End Sem Exam**	**Duration Of End Sem Exam**	**Term Work**	**Pract.**	**Oral**	**Total**
Test	**Continuous Evaluation**	**Attendance**	**Total Internal**						
15	**10**	**05**	**30**	**70**	**3 Hours**	**-**	**-**	**-**	**100**

Course outcome

1. To provide hands-on experience in designing and creating relational databases using SQL.
2. To implement various SQL queries including DDL, DML, DCL, and TCL commands.
3. To perform database design, normalization, and basic transaction management using a DBMS tool.
4. To implement fundamental data structures such as arrays, linked lists, stacks, and queues.
5. To develop programs for searching, sorting, and recursion techniques.
6. To design and implement efficient algorithms using appropriate data structures and database concepts to solve real-world computing problems.

Course Objectives

1. Create and manage databases and tables using SQL commands.
2. Execute complex SQL queries involving joins, subqueries, aggregate functions, and constraints.
3. Design normalized database schemas and implement integrity constraints in practical applications.
4. Implement linear and non-linear data structures using appropriate programming techniques.
5. Analyze and compare the performance of different sorting and searching algorithms.
6. Design and develop database-driven applications by integrating SQL programming, normalized database design, and efficient data structures to solve real-world computing problems.

Detailed Syllabus

Module / Unit	Course Module / Contents		Hours	Marks Weightage
Section I (DBMS Lab)				
	Module I			
1	1.1	Create the following table Employee with fields EmpID, E_NAME, SALARY, JOB,DB,DOJ. a. Find all the Employees who has DOB between 1 Jan 1996 to 1 Jan 2000. b. Find all the distinct DeptID in Employee table and department table.	**05**	**10%**
	1.2	a. Find name of all employees who's working in DeptID 2. b. Find all the employees that are not associated with DeptID 2 and 3		
	1.3	a. Find total number of employees and total number of departments. b. Find different number of jobs available.		
	1.4	a. Find the details of Employees who's not working as Software Engineer. b. Find the ID of those Employees whose job is Software Engineer and Salary is greater than 100000.		
	Module II:			
2	2.1	a. Hike salary of employee by 20% with position Software Engineer. b. Change DeptID to 3 whose job is Software Engineer and Salary > 50000. c. Change EmpName to xyz where EmpID is 2304 and salary not equal to 30000.	**05**	**08%**
	2.2	a. Find EmpID where salary is highest. b. Find second highest salary. c. Find EmpID of employees with second highest salary. d. d. Decrease the salary by 10% of employee who is getting second highest salary.		
	2.3	a. Select all employees that are not associated with any department. b. b. Select all departments with no employees. c. Find the names of employee working in department IT.		
	2.4	a. Create the following(s) table Salespeople with fields snum, sname, city, commission b. Orders table with fields onum, odate, snum, amt Customers table with fields cnum, cname, city, rating, snum		

3	**Module III:**		**05**	**08%**
	3.1	Create the following(s)table Salespeople with fields snum, sname, city, commission Orders table with fields onum, odate, snum, amt Customers table with fields cnum, cname, city, rating, snum a. Display name & city of salesman where city is"Pune". b. b. Display the numbers of sales persons, with orders currently in the orders table without any repeats.		
	3.2	a. List all customers not having city "Pune" or rating more than100 b. Display all customers excluding those, with rating less than equal to 100, unless they are located in "Nagar". c. Assume each salesperson has a 12% commission on order amt. d. Display orderno, snum, commission for that order. e. Display the count of snum in order table without duplication of snum.		
	3.3	1. Create a table with the under-mentioned structure (Table name isEmp) EmpNo NUMBER(4) DeptNo NUMBER(2) EmpName CHAR(10) Job CHAR(10) Manager NUMBER(4) HireDate DATE Salary NUMBER(7,2) Commission NUMBER (7,2). 2. Create a table with the under Mentioned structure (Table name is Dept) DeptNo NUMBER(2) DeptName CHAR(12) Location CHAR(12)		
		Write SQL statements to list all employees in the following format: EMPLOYEE WORKS IN DEPARTMENT Dept.No SMITH WORKS IN DEPARTMENT 20 SUDHIR WORKS IN DEPARTMENT 20 RAJWORKS IN DEPARTMENT 10 SMITHS WORKS IN DEPARTMENT 30 SANTOSH WORKS INDEPARTMENT 30		
4	**Module IV:**		**05**	**08%**
	4.1	Create a table with the under-mentioned structure (Table name is Sports) a. Student(Student No., Class, Name, Game1, Grade1, Game2, Grade2). Display the names of the students who are getting a grade C in either GAME1 orGAME2. b. Display the number of students getting grade A in cricket		
	4.2	a. Display the names of the students who have same game for both GAME1 andGAME2. b. Display the games taken by the students, whose name starts with "A". c. Assign a value 200 for Marks for all those who are getting grade B or above in GAME.		
	4.3	a. Write a command to delete those records from the table which have Grade1 ="A" and then COMMIT the changes in the database. b. Undo the task using rollback. Write a SQL command where you plan to delete the three different records from the Student table. You want to create a SAVEPOINT before each delete, so that you can ROLLBACK to any SAVEPOINT at any time to return the appropriate data to its original state		

	4.4	PL/SQL Queries: a. Write a procedure to display a message. b. Write a procedure that taken input from user and display it. c. Write a PL/SQL code to print sum of two numbers taken from the user. Write a procedure to use for loop to insert ten rows into a database table. a. Given Student Report Database, in which student marks assessment is recorded. In such schema, create a trigger so that the total and average of specified marks is automatically inserted whenever a record is insert. b. Create a trigger to calculate the percentage of the student as soon as his details are updated to the database. c. Drop an existing trigger.		
5	**Module V:**		**05**	**08%**
	5.1	a. Write a sequence query creating sequence in ascending order. b. Write a sequence query creating sequence in descending order. c. Using sequence create a table named students with columns as id and name.		
6	**Module VI:**			
	6.1	DBA Commands: a. Users, Roles and Privileges. b. Creating a User c. Password Management. d. Grant and Revoke.	**03**	**08%**
Section II (Data Structure Lab)				
	Module I		**07**	**10%**
1	1.1	a. Write a program to search an element in a given array using linear search. b. Write a program to search an element in a given array using binary search.		
	1.2	Write a program to implement a Stack, show overflow and underflow while performing push and pop operations respectively.		
	1.3	Write a program to evaluate a postfix expression.		
	1.4	Write a program to implement a queue and show the following: insertion and deletion.		
	Module II		**05**	**10%**
2	2.1	Write a program to implement a circular queue and show the following: insertion and deletion.		
	2.2	Write a program to implement Linear Linked List and show the following operations: creation, display, insertion, deletion and searching.		
	2.3	Write a program to implement Doubly Linked List and show the following operations: creation, display, insertion, deletion and searching.		
	2.4	Write a program to implement Binary tree and display the contents using preorder, post order and in order traversal techniques.		
3	**Module III**		***05***	

	3.1	Write a program to construct a Binary Search tree and perform the following operations: Insertion and Deletion of anode		
	3.2	Write a program to construct an AVL tree and perform the following operations: insertion, deletion and searching.		**10%**
	3.3	Write a program to construct an AVL tree and perform the following operations: insertion, deletion and searching.		
4	**Module IV**			
	4.1	Write a program to implement Depth First Search using linked representation of graph.	**05**	**10%**
	4.2	Write a program to implement Breadth First Search using linked representation of graph.		
5	**Module V**			
	5.1	Write a program to create a minimum spanning tree using Kruskal "algorithm. Write a program to create a minimum spanning tree using Prim "algorithm.	**02**	**10%**
	5.2	Write a program to sort a given set of elements using Selection sort and hence find the time required to sort elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n.		
Total			**52**	**100%**

Recommended Books:

No	Book Title	Author(s)	Publisher	ISBN-13 / Edition
1	Fundamentals of Database Systems	Ramez Elmasri & Shamkant B. Navathe	Addison-Wesley	978-0133970777 (7th Ed.)
2	Database System Concepts	Abraham Silberschatz, Henry F. Korth, & S. Sudarshan	McGraw-Hill	978-0073523323 (7th Ed.)
3	Database Management Systems	A.K. Majumdar & P. Bhattacharya	Tata McGraw-Hill	978-0074622308 (1st Ed.)
4	An Introduction to Database Systems	C.J. Date	Addison-Wesley	978-0321197849 (8th Ed.)
5	Mastering Data Structures Through C Language	J.B. Dixit	Laxmi Publications	978-8131804476 (1st Ed.)
6	Data Structures Using C	Yashavant Kanetkar	BPB Publications	978-8183334617 (2nd Ed.)
7	Introduction to Algorithms	T. H. Cormen, C. E. Leiserson, R. L. Rivest, & C. Stein	MIT Press / PHI	978-0262046305 (4th Ed.)